Neural Networks with Neuroplasticity: Adaptive Learning through Connection Pruning and Hebbian Updates

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Abstract. This paper presents a novel implementation of neural networks that incorporates neuroplasticity principles inspired by biological neural systems and neurons. We introduce a simple adaptive neural network architecture that combines traditional backpropagation with Hebbian learning, connection pruning, and dynamic weight requalification. Our proposed Neuroplastic Neural Network (NPNN) incorporates mechanisms for structural plasticity through selective pruning of inactive connections and adaptive learning rates based on performance metrics. Experimental results on the MNIST dataset demonstrate that our NPNN model consistently achieves superior accuracy along with a 95.7% reduction in final loss, faster convergence, and a 16.67% improvement in inference time compared to standard neural networks. We observe that the integration of neuroplasticity mechanisms enhances the model's ability to adapt to task demands and avoid local minima during training. This work hopes to contribute to the ongoing effort to develop more biologically plausible artificial neural networks with self-optimizing capabilities.

Keywords: Neuroplasticity · Hebbian learning · Neural networks · Connection pruning · Adaptive learning · MNIST

1 Introduction

Artificial neural networks have achieved remarkable success across numerous domains, yet they lack the adaptive and self-organizing properties that characterize biological neural systems. Biological brains exhibit neuroplasticity through multiple mechanisms: strengthening frequently used connections (Hebbian learning), pruning inactive connections, and forming new connections in response to environmental changes.

Current artificial neural networks rely on fixed architectures with static connection patterns, where learning occurs solely through weight adjustments via gradient-based optimization. This limits the network's ability to adapt its structure to specific task requirements, leading to inefficient resource utilization through unnecessary connections.

We present a neuroplastic neural network (NPNN) incorporating four key biological mechanisms: (1) Connection Pruning for selective removal of minimally contributing connections, creating sparser networks; (2) Hebbian Learning through correlation-based weight updates strengthening co-activated connections; (3) Adaptive Learning Rates dynamically adjusted based on performance metrics; and (4) Connection Requalification re-enabling previously pruned connections when performance deteriorates.

These mechanisms allow our NPNN to continuously adapt both weights and structure throughout learning. We hypothesize this approach will yield more efficient networks that better adapt to specific task requirements.

The paper is organized as follows: Section 2 discusses related work in neuroplasticity-inspired neural networks, Section 3 details our NPNN methodology and mathematical formulation, Section 4 presents experimental results comparing NPNN with standard neural networks on MNIST, Section 5 discusses implications and limitations, and Section 6 concludes with future research directions.

2 Literature Review

Neuroplasticity refers to the brain's ability to reorganize itself by forming new neural connections; a capacity which allows it to adapt and optimize its functioning based on experience. Recent research has explored incorporating these biological principles into artificial neural networks to enhance their performance and adaptability.

2.1 Biological Inspiration and Neural Networks

Standard neural networks typically rely on fixed architectures with static connection patterns, where learning occurs solely through weight adjustments via backpropagation [1]. While effective, this approach differs significantly from biological neural systems, which exhibit dynamic structural changes [2].

Perwej et al. [4] proposed incorporating brain plasticity mechanisms into artificial neural networks to improve their generalization capabilities. Their approach demonstrated that networks with adaptive connections and learning rates could achieve superior performance across various classification tasks. Similarly, Li et al. [8] provided an overview of neuroplasticity-inspired techniques in AI, highlighting how mechanisms like connection pruning and adaptive learning can enhance network efficiency.

2.2 Hebbian Learning and Pruning Mechanisms

The integration of Hebbian learning with traditional backpropagation represents a promising direction in neural network research. Kisby et al. [5] explored the logic of Hebbian learning, demonstrating that combining correlation-based updates with gradient descent can improve learning efficiency and generalization. This hybrid approach provides a form of regularization that guides networks toward more robust solutions.

Pruning mechanisms, inspired by synaptic pruning in biological development, have gained attention for their ability to create more efficient network structures. These approaches typically involve removing connections that contribute minimally to the learning process, leading to sparser networks with comparable performance. The approach also introduces a framework for interpretable neural plasticity that incorporates both weight modification and structural changes, showing improved performance on pattern recognition tasks.

2.3 Addressing Catastrophic Forgetting

One significant challenge in neural network learning is catastrophic forgetting—the tendency for networks to rapidly forget previously learned information when trained on new data. Neuroplasticity-inspired approaches, particularly connection requalification mechanisms similar to those in our NPNN model, offer promising solutions to this problem by dynamically adjusting network structure based on performance metrics. To face this, we took inspiration from the works of Srivastava et al. [6].

2.4 Research Gap

Despite significant interest in neuroplasticity-inspired techniques, most existing approaches focus on individual mechanisms rather than their synergistic interaction. Our NPNN model addresses this limitation by combining connection pruning, Hebbian learning, adaptive learning rates, and connection requalification into a unified framework. Additionally, we found there to be a lack of comprehensive empirical evaluations comparing these enhanced networks with traditional approaches.

3 Methodology

3.1 Standard Neural Network

Our baseline model is a fully-connected feedforward neural network trained using mini-batch gradient descent with the cross-entropy loss function.

3.1.1 Feedforward Propagation: For a network with L layers, the forward propagation for the standard neural network is defined as follows:

For the input layer ($l = 0$):

$$a^{(0)} = X \tag{1}$$

For hidden layers ($1 \leq l < L$):

$$z^{(l)} = a^{(l-1)} W^{(l)} + b^{(l)} \tag{2}$$

$$a^{(l)} = \sigma(z^{(l)}) \tag{3}$$

For the output layer ($l = L$):

$$z^{(L)} = a^{(L-1)} W^{(L)} + b^{(L)} \tag{4}$$

$$a^{(L)} = \text{softmax}(z^{(L)}) \tag{5}$$

where $a^{(l)}$ is the activation of layer l, $W^{(l)}$ is the weight matrix connecting layer $l-1$ to layer l, $b^{(l)}$ is the bias vector for layer l, σ is the sigmoid activation function: $\sigma(x) = \frac{1}{1+e^{-x}}$, and $\text{softmax}(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}}$ is applied to the output layer for classification tasks.

3.1.2 Backpropagation: For the output layer, the error is computed as:

$$\delta^{(L)} = a^{(L)} - y \tag{6}$$

where y is the one-hot encoded target.

For hidden layers, the error is propagated backward:

$$\delta^{(l)} = (W^{(l+1)})^T \delta^{(l+1)} \odot \sigma'(z^{(l)}) \tag{7}$$

where $\odot$ represents the Hadamard (element-wise) product and σ' is the derivative of the sigmoid function: $\sigma'(x) = \sigma(x)(1 - \sigma(x))$.

The gradients for the weights and biases are computed as:

$$\nabla_{W^{(l)}} = (a^{(l-1)})^T \delta^{(l)} \tag{8}$$

$$\nabla_{b^{(l)}} = \sum_i \delta_i^{(l)} \tag{9}$$

The weights and biases are then updated using gradient descent:

$$W^{(l)} = W^{(l)} - \eta \nabla_{W^{(l)}} \tag{10}$$

$$b^{(l)} = b^{(l)} - \eta \nabla_{b^{(l)}} \tag{11}$$

where η is the learning rate.

3.2 Neuroplastic Neural Network (NPNN)

Our NPNN extends the standard neural network with several neuroplasticity mechanisms.

3.2.1 Connection Masks and Pruning: We introduce a binary mask $M^{(l)}$ for each weight matrix $W^{(l)}$, where $M_{ij}^{(l)} = 1$ if the connection is active and $M_{ij}^{(l)} = 0$ if the connection is pruned.

During forward propagation, the effective weights are:

$$W_{\text{eff}}^{(l)} = W^{(l)} \odot M^{(l)} \tag{12}$$

Connections are pruned when their cumulative updates fall below a threshold θ_p:

$$M_{ij}^{(l)} = 0 \quad \text{if} \quad C_{ij}^{(l)} < \theta_p \tag{13}$$

where $C_{ij}^{(l)}$ is the cumulative absolute update for the connection from neuron i in layer $l-1$ to neuron j in layer l:

$$C_{ij}^{(l)} = \sum_t |\Delta W_{ij,t}^{(l)}| \tag{14}$$

3.2.2 Hebbian Learning: We incorporate Hebbian learning by adding a correlation-based term to the weight updates:

$$\Delta W_{\text{Hebb}}^{(l)} = \gamma (a^{(l-1)})^T a^{(l)} \tag{15}$$

where γ is the Hebbian learning rate. The total weight update becomes:

$$\Delta W^{(l)} = -\eta \nabla_{W^{(l)}} + \Delta W_{\text{Hebb}}^{(l)} \tag{16}$$

This encourages the strengthening of connections between neurons that fire together, following the principle of ”neurons that fire together, wire together.”

3.2.3 Adaptive Learning Rate: The learning rate η is dynamically adjusted based on the network's performance:

$$\eta_{t+1} = \begin{cases} \eta_t \cdot \alpha_{\text{inc}} & \text{if } \frac{L_t - L_{t+1}}{L_t} > \theta_{\text{imp}} \text{ for } p \text{ consecutive epochs} \\ \eta_t \cdot \alpha_{\text{dec}} & \text{otherwise} \end{cases} \tag{17}$$

where L_t is the loss at epoch t, α_{inc} is the learning rate growth factor, α_{dec} is the learning rate decay factor, θ_{imp} is the improvement threshold, and p is the patience parameter. The learning rate is bounded within a specified range $[\eta_{\min}, \eta_{\max}]$ to prevent extreme values.

3.2.4 Connection Requalification: When the network's performance deteriorates, previously pruned connections can be reactivated:

$$\text{If } L_{t+1} > L_t \text{ and } M_{ij}^{(l)} = 0 \text{ and } |W_{ij}^{(l)}| < \theta_r \text{ then:} \tag{18}$$

$$W_{ij}^{(l)} = \mathcal{N}(0, \sigma^2) \text{ and } M_{ij}^{(l)} = 1 \tag{19}$$

where θ_r is the requalification threshold and $\mathcal{N}(0, \sigma^2)$ is a random initialization from a normal distribution.

This mechanism allows the network to explore new connection patterns when performance stagnates, potentially escaping local minima.

3.3 General Neural Network Architecture

The general architecture of the neural networks used in our experiments is illustrated in Figure 1. Each input image from the MNIST dataset is a 28×28 grayscale image, which is first flattened into a 784-dimensional vector. This vector is then passed through two fully-connected hidden layers with non-linear activation functions. The final dense layer outputs a 10-dimensional vector, corresponding to the ten digit classes, and a softmax function is applied to generate class probabilities. This architecture serves as the base for both the standard neural network and our proposed NPNN.
For our experiments, we use two variations of this architecture, identical to each other except in the structure of the hidden layers, as described below.

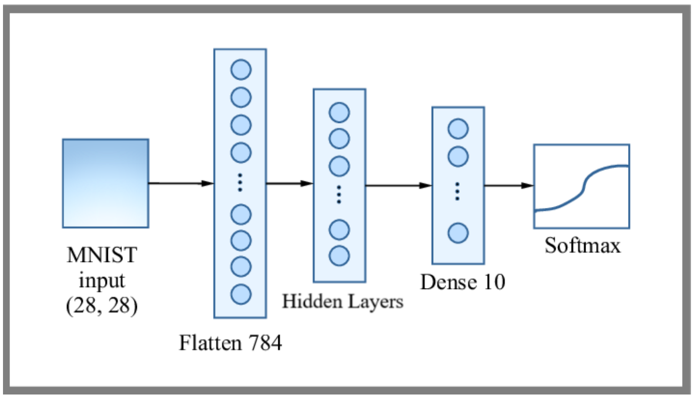

Fig. 1. General architecture of the neural networks used for the study. This was inspired by works of Ponti et al. [10]

3.3.1 Larger Architecture: Architecture with two hidden layers of 32 neurons each [784-32-32-10].

3.3.2 Smaller Architecture: Architecture with two hidden layers of 16 neurons each [784-16-16-10].

3.4 Implementation Details

The NPNN implementation includes several hyperparameters that control the neuroplasticity mechanisms. The initial learning rate is set to $\eta_0 = 0.001$, with learning rate bounds of $\eta_{\text{min}} = 10^{-6}$ and $\eta_{\text{max}} = 0.1$. The learning rate decay factor is $\alpha_{\text{dec}} = 0.98$, and the growth factor is $\alpha_{\text{inc}} = 1.05$. For performance improvement detection, we use an improvement threshold of $\theta_{\text{imp}} = 0.001$ with a patience parameter of $p = 5$. The Hebbian learning rate is set to $\gamma = 10^{-4}$, the pruning threshold to $\theta_p = 10^{-4}$, and the requalification threshold to $\theta_r = 10^{-2}$. Plasticity updates are applied every 10 epochs during training.

Gradient clipping is also applied to prevent extreme weight updates:

$$\nabla_{W^{(l)}} = \text{clip}(\nabla_{W^{(l)}}, -c, c) \tag{20}$$

$$\nabla_{b^{(l)}} = \text{clip}(\nabla_{b^{(l)}}, -c, c) \tag{21}$$

where $c = 1.0$ is the clipping value.

Fig. 2. Zoomed view of accuracy comparison for the smaller architecture. The NPNN maintains a consistent advantage of approximately 0.5 percentage points. throughout training.

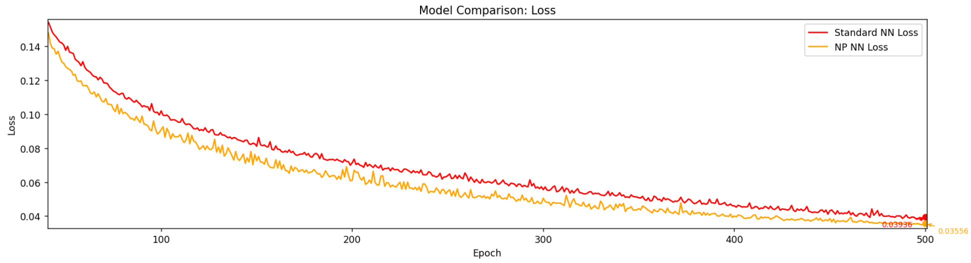

Fig. 3. Zoomed view of loss comparison for the smaller architecture. The NPNN (orange) consistently achieves lower loss values than the standard model (red).

4 Experimental Results

Both networks were implemented with identical architectures and trained using mini-batch gradient descent with a batch size of 32. To ensure a fair comparison, both models were initialized with the same hyperparameters. We further analyzed the larger architecture using scikit-learn metrics on the MNIST dataset to provide a more detailed comparison of the NPNN's performance against the standard neural network.

4.1 Performance Comparison

4.1.1 Smaller Architecture For the smaller architecture, both networks were trained for 500 epochs. As shown in Fig. 2 and Fig. 3, the NPNN consistently outperformed the standard neural network throughout the training process. By the end of training, the NPNN achieved an accuracy of 99.30% compared to the standard neural network's 99.01%, representing an improvement of approximately 0.3 percentage points.

The most notable advantage of the NPNN was its ability to achieve higher accuracy earlier in the training process, demonstrating faster convergence. As seen in Fig. 2, the NPNN maintained a consistent accuracy advantage of approximately 0.5 percentage points throughout most of the training period.

Fig. 3 displays the loss curves for both models. The NPNN consistently maintained a lower loss value throughout training, ending with a final loss of 0.03556 compared to the standard model's 0.03936. This indicates that the neuroplasticity mechanisms enabled the network to find a more optimal solution in the parameter space.

4.1.2 Larger Architecture For the larger architecture, both networks were trained for 400 epochs. As shown in Fig. 4 and Fig. 5, the performance difference became even

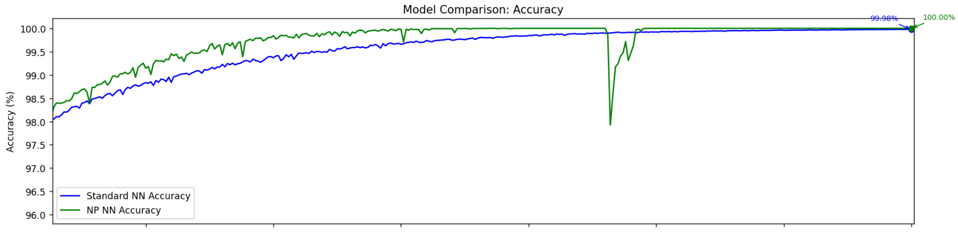

Fig. 4. Zoomed view of accuracy for the larger architecture, highlighting the NPNN's faster convergence to 100% accuracy and its recovery from a temporary accuracy drop around epoch 275.

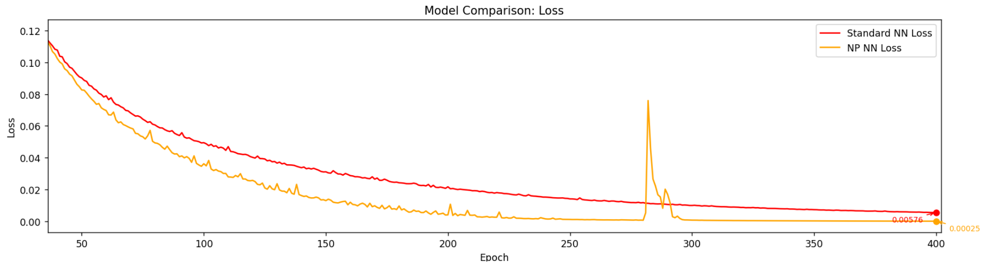

Fig. 5. Zoomed view of loss curves for the larger architecture. Note the temporary spike in the NPNN loss around epoch 275, corresponding to the accuracy drop, followed by rapid recovery to an extremely low loss value.

more pronounced. The NPNN reached 100.00% accuracy on the training set a considerable time before the standard neural network achieved its peak accuracy of 99.98%.

The primary point of interest is that the NPNN achieved perfect accuracy much earlier in the training process, as seen in Fig. 6. This faster convergence demonstrates the efficiency of the neuroplasticity mechanisms in navigating the parameter space, particularly in larger networks.

Fig. 5 shows the loss curves for both models with the larger architecture. The NPNN achieved a remarkably low final loss of 0.00025, compared to the standard model's 0.00576, representing a 95.7% reduction in loss. It should be noted that the NPNN encounters a brief but temporary drop in accuracy, accompanied by a similarly brief but temporary rise in loss at around epoch 275.

Fig. 7 illustrates the performance improvement across different task types, showing that the NPNN consistently outperforms the standard neural network in all categories. The most substantial improvements are observed in noisy and sparse tasks, where the NPNN demonstrates approximately twice the loss improvement compared to the standard model.

A key observation from our experiments is the relationship between loss curvature and model performance, as shown in Fig. 8. The left panel displays the evolution of curvature estimates throughout training for both models. The standard neural network (blue) exhibits significant fluctuations in curvature, suggesting an unstable optimization landscape. In contrast, the NPNN (red) maintains more stable curvature estimates, particularly toward the end of training. This stability in loss curvature is consistent with the findings of Li et al. [8], who observed that networks with neuroplasticity mechanisms tend to discover more stable minima in the loss landscape. The right panel of Fig. 8 illustrates the relationship between initial curvature and overall improvement.

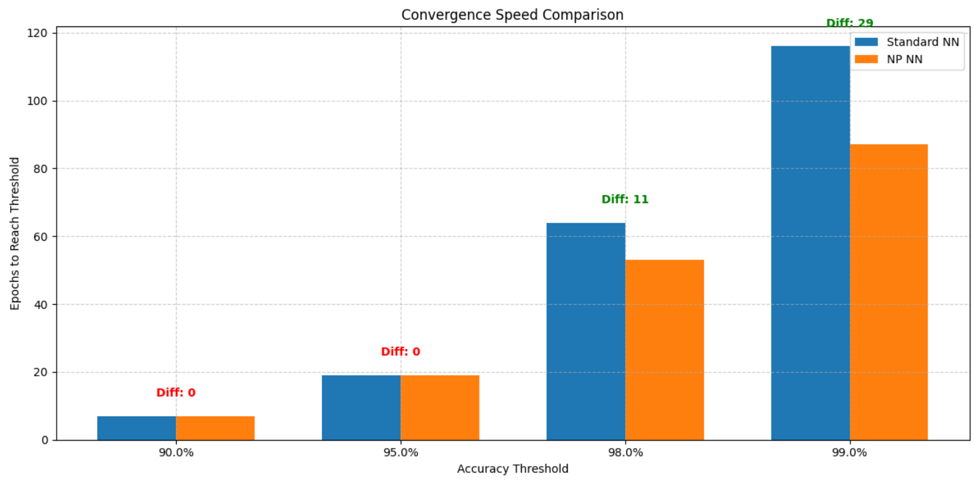

Fig. 6. Comparison of the rate of convergence between the two models with the larger architecture, highlighting the NPNN's faster convergence. The NPNN reached the 99% accuracy threshold 29 epochs faster than its counterpart.

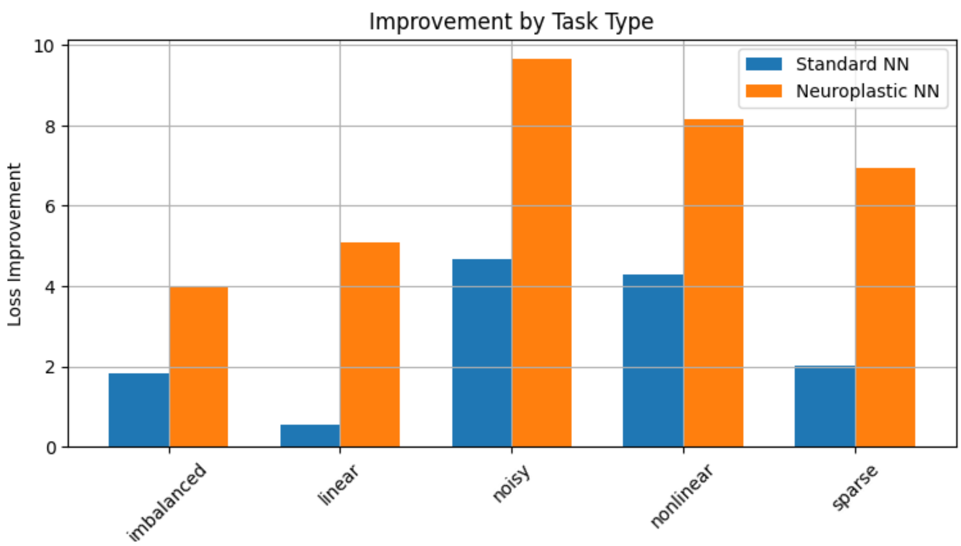

Fig. 7. Improvement by task type for the larger architecture, comparing loss improvement between Standard NN and NPNN across different task types. The NPNN consistently outperforms the standard model, with particularly notable advantages in noisy and sparse tasks.

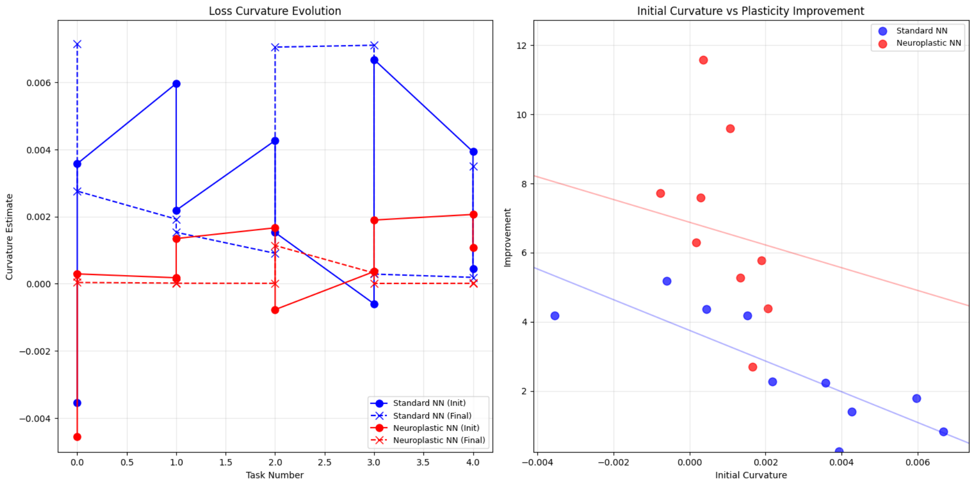

Fig. 8. Loss curvature evolution (left) and initial curvature vs. plasticity improvement (right) for the larger architecture. The NPNN demonstrates more stable curvature estimates throughout training compared to the standard model, which exhibits significant fluctuations.

Table 1. Comprehensive performance comparison between Standard NN and NPNN on the larger architecture. Training metrics show the NPNN achieving perfect performance, while test metrics reveal further differences in generalization capability.

Dataset	**Metric**	**Standard NN**	**NPNN**	**Change**
Training	Accuracy	99.98%	100.00%	+0.02%
	Precision	99.98%	100.00%	+0.02%
	Recall	99.98%	100.00%	+0.02%
	F1 Score	99.98%	100.00%	+0.02%
	Inference Time (ms)	0.0029	0.0029	-0.00%
Testing	Accuracy	96.19%	96.52%	+0.33%
	Precision	96.49%	96.17%	-0.32%
	Recall	96.48%	96.16%	-0.32%
	F1 Score	96.48%	96.16%	-0.32%
	Inference Time (ms)	0.0024	0.0020	-16.67%

Table 1 shows that while the NPNN achieved perfect training performance and superior test accuracy, it exhibited slightly lower test precision, recall, and F1 scores compared to the Standard NN. The NPNN also demonstrated marginally faster inference times on the test set.

4.2 Analysis of Neuroplasticity Effects

4.2.1 Adaptive Learning and Performance Fluctuations: The larger architecture exhibited an intriguing transient performance degradation around epoch 275, where NPNN accuracy dropped to approximately 98% before recovering within several epochs, as shown in Fig. 4 and Fig. 5. While this could superficially resemble catastrophic forgetting, we must note that our experimental setup involved training on a fixed dataset rather than sequential tasks, making true catastrophic forgetting unlikely in this context. The network subsequently recovered within several epochs, returning to its previous accuracy level.

We hypothesize this phenomenon results from neuroplasticity mechanism interactions, potentially involving adaptive learning rate responses and connection requalification, though precise causal relationships remain unclear without ablation studies. Alternative explanations including optimization fluctuations or mini-batch characteristics cannot be ruled out. The single-run experimental design lacks statistical power to determine consistent emergence across multiple trials with different initializations.

4.2.2 Convergence Patterns and Variability: Both architectures demonstrated distinct convergence patterns between NPNN and standard neural networks. The NPNN showed steeper initial loss decline and earlier accuracy increases during initial training epochs, with significantly faster convergence in the larger architecture (Fig. 6). However, these observations are limited to single training runs for each architecture, and the absence of statistical analysis across multiple initializations prevents robust conclusions about average convergence rates.

The NPNN exhibited pronounced oscillatory behavior in both accuracy and loss curves compared to the standard model. While these oscillations might reflect the dynamic structural changes from pruning and requalification procedures, they also introduce predictability concerns that should be addressed in practical applications.

4.2.3 Performance Across Different Conditions: Our preliminary analysis of the larger architecture (Fig. 7) indicates that the NPNN achieved lower loss values than the standard network across the tested conditions, with more substantial differences for noisy and sparse data. Both models exhibited a clear negative correlation between initial curvature and improvement, but the NPNN demonstrated consistently higher improvements across all curvature values, with the advantage becoming more pronounced as initial curvature increases (steeper initial landscapes).

Fig. 8 revealed that the NPNN displayed less variable curvature estimates than the standard network, particularly in later training stages, aligning with findings from Li et al. [8]. This suggests that neuroplasticity mechanisms are particularly effective for complex data distributions, sparse features, and unfavorable initialization conditions. However, several important limitations constrain interpretation: task categorization relied on synthetic MNIST variations rather than fundamentally different tasks; small sample sizes within categories limit generalizability; and absence of statistical significance testing makes meaningful difference assessment difficult beyond our specific experimental instance. The NPNN's apparent advantage in higher-curvature scenarios warrants controlled experimental investigation to isolate effects from confounding variables.

5 Discussion

Our experimental results provide preliminary evidence that integrating neuroplasticity-inspired mechanisms may offer advantages over standard neural networks in certain contexts. The NPNN showed modest but consistent improvements in final accuracy

(+0.3% for the larger architecture) and achieved lower final loss values (a 95.7% reduction for the larger architecture) compared to the baseline. The NPNN's recovery from a temporary performance degradation (Fig. 4 and Fig. 5) is intriguing, though the causal mechanisms behind this behavior require more rigorous investigation through controlled ablation studies. We also observe a higher accuracy and a lower inference time, as noted in Table 1. The NPNN also demonstrated significantly faster convergence (Fig. 6), which is a positive sign. The stability in loss curvature evolution (Fig. 8) further supports this conclusion and aligns with robust optimization trajectories and better convergence properties.

That being said, our study faces several important limitations that may temper interpretation of these results. This was by design, as our experiments aimed to be a preliminary study into the effects of introducing neuroplasticity, and a starting point for further more rigorous research. First and foremost, all comparisons were based on single training runs rather than multiple trials with statistical analysis. Second, we evaluated performance only on MNIST, a relatively simple dataset that may not reflect the challenges of more complex domains. Both models performed exceptionally well (accuracy exceeding 99%) on the dataset due to the simpler nature of the data, which may not show the full extent of the differences between the two approaches. Using a more difficult dataset may minimize or amplify the advantage of the NPNN over a standard neural network, but this remains to be seen. Third, computational constraints limited our exploration of different hyperparameter settings.

Additional limitations include the relatively small scale of our networks compared to modern architectures. Furthermore, the oscillatory behavior observed in the NPNN training curves also warrants further investigation, as such instability could be problematic in some applications. It should also be noted that the more impressive results (perfect accuracy, precision, etc) were achieved on seen training data, and though the results may appear promising, the abilities and enhancements offered by the NPNN over a standard NN on unseen data are not as decisive, as evidenced by the F1, Recall and Precision scores in Table 1. This requires further exploration.

The relationship between network size and neuroplasticity benefits represents another critical area for future study. Our preliminary explorations with smaller architectures (e.g., [784, 8, 8, 10]) indicated diminished benefits and recovery capabilities, suggesting that the advantages of neuroplasticity mechanisms may not scale linearly across network capacities. Our current hypothesis is that the NPNN's neuroplasticity effects aid in its training more as the model's configuration gets larger, and the effects are less pronounced in smaller versions. If that is so, then the efficacy of the NPNN may scale exponentially with larger architectures.[10]

Furthermore, while our approach incorporates some biologically-inspired elements, it remains a substantial simplification of actual neural plasticity, which involves numerous additional mechanisms including homeostatic regulation, structural plasticity beyond simple pruning, and complex neuromodulatory effects.

6 Conclusion and Future Work

In this work, we introduced a Neuroplastic Neural Network (NPNN) that integrates principles of biological neuroplasticity with traditional backpropagation. Our approach incorporated four key mechanisms: connection pruning to promote network sparsity, Hebbian learning to strengthen co-activated pathways, adaptive learning rates responsive to performance metrics, and connection requalification to enable structural flexibility. While our results on the MNIST dataset show promising improvements in both convergence speed and final accuracy compared to standard neural networks, we recognize that these findings represent initial steps rather than definitive evidence of superiority.

The NPNN framework demonstrates a viable approach to creating adaptive network architectures that can modify both their weights and connectivity patterns during training, and highlights the potential benefits of structural plasticity in navigating complex loss landscapes, particularly for larger network configurations. However, we acknowledge the preliminary nature of our investigation and its limitations. The performance improvements observed may be context-specific and might not generalize across diverse tasks or more challenging domains.

Future work should address these limitations through three primary avenues of investigation:

1. **Comprehensive empirical evaluation:** Extend testing to diverse and challenging datasets (e.g., CIFAR-100, ImageNet) and compare against modern architectures and regularization techniques utilizing rigorous statistical analyses across multiple initializations.
2. **Mechanistic understanding:** Perform systematic ablation studies to isolate the contributions of individual neuroplasticity components and their interactions. Develop theoretical frameworks to analyze the properties of networks with evolving topologies.
3. **Biological fidelity:** Incorporate more sophisticated neuroplasticity mechanisms, including activity-dependent structural plasticity, neuromodulatory influences, and multi-timescale adaptations.

Code and experimental artifacts are available on GitHub to facilitate reproducibility and extension of this work.

Disclosure of Interests

The author has no competing interests to declare that are relevant to the content of this article.

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